Chapter 53 - Electro-Magnetic Waves
Calculus Physics II

Review of the Physics of Waves

Wave - A wave is a traveling disturbance that travels through space in such a way that energy travels through the medium, but matter does not.

- A wave carries energy from place to place.
- It takes work to create a wave therefore the wave has energy and can do work.

Types of waves

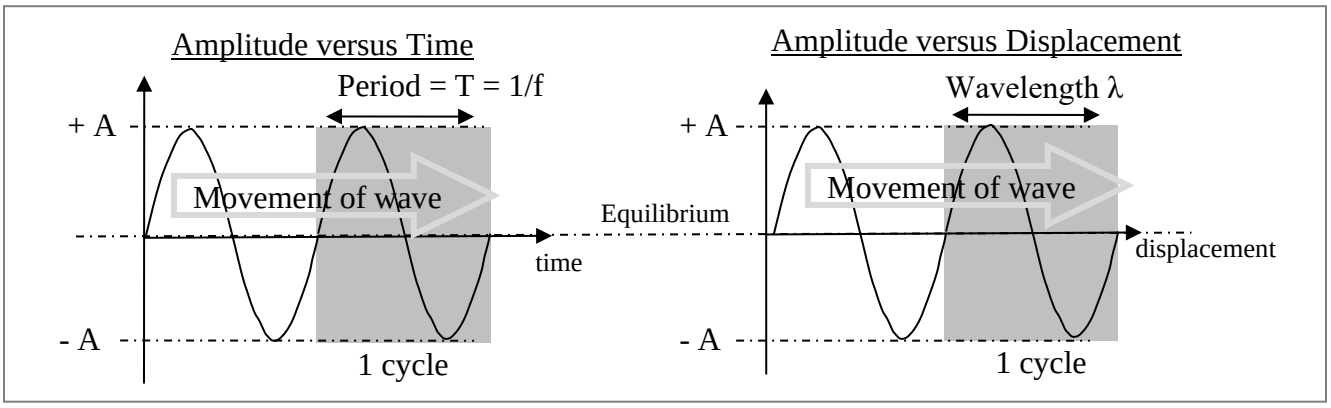
Mechanical waves – waves that travel through a substance, medium or material. They must have a medium to travel through.

- Mechanical waves use the elastic properties of the medium to move the wave along.
- They are organized disturbance of the particles in the medium.
- The energy of the wave is in the motion of the particles in the medium.
  Examples: sound waves, water waves, waves on a string.

Electromagnetic waves - is a self-propagating wave made up of the electric and magnetic fields.

- No medium needed to propagate (we can see the sun and stars in outer space through vacuum).
- The energy in the wave is in the electric and magnetic fields.
  Examples: Radio waves, microwaves, infrared radiation, visible light, ultraviolet, x-rays, gamma rays.

Wave properties – Period (T), wavelength (λ), frequency (f), amplitude (A) and speed (v) describe the properties of a wave.



Amplitude (A) – Maximum displacement from equilibrium	(units: meters)
Period (T) - time to complete one cycle. T = 1/f	(units: seconds)
Wavelength (λ) – distance traveled in one cycle	(units: meters)
Frequency (f) – # of cycles per second f = 1/T	(units: Hz or cycles/s)
Angular velocity $\omega = 2\pi/T = 2\pi f$	(units: radians/s)

Wave velocity (v_{wave}) – is the propagation speed of the wave, the magnitude of the velocity of a wave "bump" . It is how fast the wave moves through the medium. The wave velocity is equal to the frequency (f) of the wave times its wavelength (λ).

$$v_{wave} = f\lambda$$ Wave equation Units of m/s

Electromagnetic Waves (or EM Radiation)– EM waves are made up of fluctuating electric and magnetic fields created by accelerating charge and propagate together through transparent medium or vacuum.

The changing magnetic field creates the electric field, and the changing electric field creates the magnetic field, the wave is self-sustaining and does not need a medium to travel through and can propagate in a vacuum.

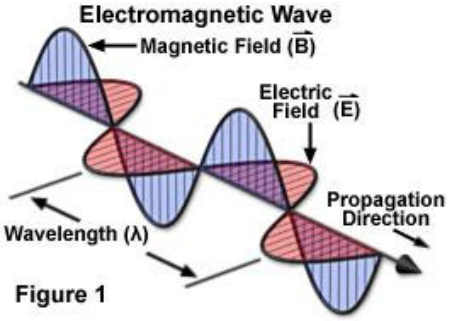


Figure 1

- EM waves travels at the same speed that the electric and magnetic fields do and that is the speed of light $c = 3 \times 10^8$ m/s or 670 million mph in vacuum.

  - Charged particles at rest or moving at a constant velocity do NOT emit EM wave.
  - Every vibrating charged object creates a disturbance (wave) in its own electromagnetic field. This disturbance is an EM wave and spreads outward at the speed of light.

"Light" is a general term for all of the electromagnetic spectrum that includes EM radiation from radio waves to gamma rays.

Visible light is made up of Electromagnetic waves in the visible part of the EM spectrum and can be detected (seen) by the human eye.

Medium – is the "stuff" the waves travel through; it is the matter or material or substance that moves when the wave passes.

Isotropic source – A source of EM waves that radiates with equal magnitude in all directions, electromagnetic radiation from a light bulb is nearly Isotropic.

Transverse wave - a wave in which the vibrations of the medium or field are at right angles to the direction of propagation. The Electromagnetic wave is a transverse wave.

James Clerk Maxwell put together four of the most important equations in modern physics and are known as "Maxwell's Equations", the equations bought together Electrostatics and Magnetism and completely describe the classical wave nature of electromagnetic waves (shown below). They also laid the groundwork for Einstein's Theory of Relativity.

Maxwell's Equations for EM waves Maxwell's Eqution relate fields to their sources (charges and currents) and describe how they vary in space and time.

Name	Differential form	Integral form
Gauss's law	$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0}$	$\oiint_{\partial V} \mathbf{E} \cdot \mathrm{d}\mathbf{A} = \frac{Q(V)}{\varepsilon_0}$
Gauss's law for magnetism	$\nabla \cdot \mathbf{B} = 0$	$\oiint_{\partial V} \mathbf{B} \cdot \mathrm{d}\mathbf{A} = 0$
Maxwell–Faraday equation (Faraday's law of induction)	$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$	$\oint_{\partial S} \mathbf{E} \cdot \mathrm{d}\mathbf{l} = -\frac{\partial \Phi_{B,S}}{\partial t}$
Ampère's circuital law (with Maxwell's correction)	$\nabla \times \mathbf{B} = \mu_0\mathbf{J} + \mu_0\varepsilon_0\frac{\partial \mathbf{E}}{\partial t}$	$\oint_{\partial S} \mathbf{B} \cdot \mathrm{d}\mathbf{l} = \mu_0 I_S + \mu_0\varepsilon_0\frac{\partial \Phi_{E,S}}{\partial t}$

Gauss' Law - states that electric charges produce electric fields. The total electric flux through a closed surface is proportional to the total electric charge enclosed within it.

- Electric charge produces Electric Field
- Electric field lines are not closed the have a beginning and end.

Gauss's Law for Magnetic Fields – The total flux through an area enclosing the charge is zero. Magnetic field lines always form continuous loops, meaning the total magnetic flux through a closed surface is always zero.

- Magnetic flied lines are closed loops, they do not have a beginning or end.

Faraday's Law of Induction - A changing magnetic field can create an electric field.

Ampere's Law – A current or a changing electric field creates a magnetic field.

- Both current and changing Electric fields are sources of magnetic fields
- Once the EM wave is generated it no longer depends on source!

Unified Theory: Maxwell's equations show that electricity and magnetism are two aspects of a single phenomenon: electromagnetism.

Maxwell's Equations determine the Nature of EM Waves – in addition to describing the properties of Electromagnetic waves, the equations set the limit for the speed of electromagnetic waves at the speed of light (c) that propagates in vacuum. The equations also show that the speed of light is a constant for the entire electromagnetic spectrum when propagating in a vacuum.

$$v_{EM\ wave} = \frac{1}{\sqrt{\epsilon_o \mu_o}} = c \qquad \text{Speed of EM waves in a vacuum}$$

$\mu_o = 1.257 \times 10^{-6}$ Tm/A $\qquad\qquad$ $\varepsilon_o = 8.85 \times 10^{-12}$ C^2/Nm^2

Speed of Light in vacuum $c = 299{,}792{,}458$ m/s $= 3.0 \times 10^8$ m/s

$= 186{,}000$ mi/s $= 670$ million mi/hr

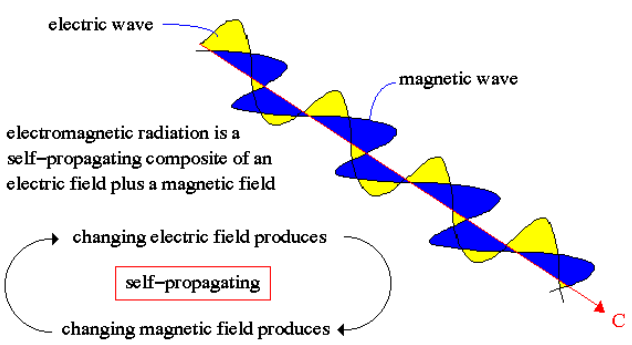
- Speed of light in vacuum is not infinite and is the absolute maximum velocity anything can travel. Nothing can go faster than the speed of light! (that we know of)
- The speed (in vacuum) of an EM wave is the same for all frequencies in the EM spectrum.
- The speed of an EM wave is the same (constant) in all reference frames.
- The speed of light depends on what it is propagating through (μ and ε).
- All fields propagate at the speed of light; electric, magnetic, and gravitational.

Example: If the wavelength of green light is 530 nm and it is traveling in a vacuum, what is the frequency of the green light?

$c = \lambda f$ therefore $f = c/\lambda = (3 \times 10^8 \text{ m/s})/(530 \times 10^{-9} \text{ m}) = 5.66 \times 10^{14}$ Hz $= 566$ THz

Propagation through vacuum – A changing electric field generates a magnetic field and a changing magnetic field generates an electric field. As the wave propagates, the changing fields generate each other as they change in magnitude.

- The energy created by the changing electric field is the magnetic field and the energy created by the changing magnetic field is the electric field of the EM wave.
- EM waves are self-sustaining waves that no longer need the source of the energy to survive.
- Energy is the same for both the electric and magnetic fields and is conserved for both Electric and Magnetic waves can go for very, very long distances (Across the universe).



Light year – is a measure of distance and it is the distance light would travel in one year.

1 light year = 5.88 trillion miles (5.88×10^{12} miles) = 9.5×10^{15} m

Example: If the sun is 93 million miles (1.5×10^{11} m) from the earth how long would it take for the EM waves from the sun to get the earth?

$v = \Delta x/\Delta t = c$ therefore $\Delta t = \Delta x/v = 500$ s = 8.3 min.

The closest star to earth is Alpha Centauri and is about 4.4 light years away how many meters is this distance and if you had a spaces ship that could travel at 25,000 mph (11,000 m/s) (maximum speed of the Apollo space craft going to the moon) how much time would it take you to get there in years? 1 year = 8760 hours

4.4 ly x (9.5×10^{15} m/ly) = 4.18×10^{16} m = 2.6×10^{13} miles = 26 trillion miles!

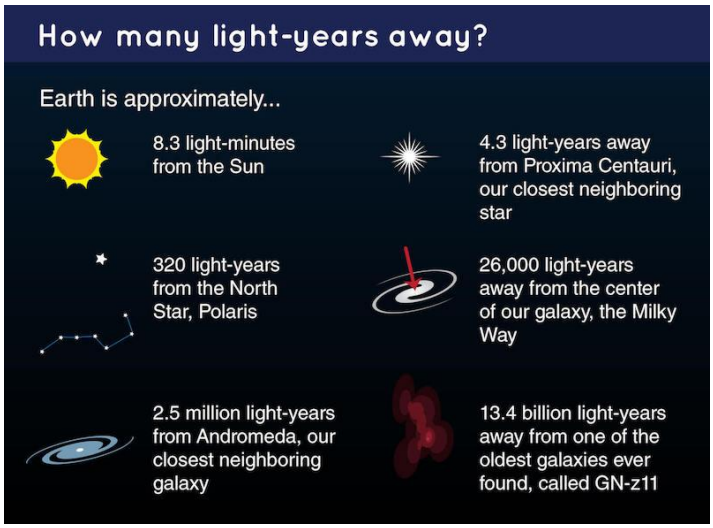
$v = \Delta x/\Delta t$ therefore $\Delta t = \Delta x/v = 1.04 \times 10^{9}$ hours = 120,000 years

For context, the Milky way Galaxy is about 200,000 ly in diameter therefore it would take about 5.4 billion years to cross the Milky way going 25,000 mph!

Seeing back in time? Because the distances are so large in our universe it takes light a long time to reach Earth from other parts of our galaxy or universe. The light that we see from stars and galaxies was emitted many, many years before we see it.

We see the sun the way it was about 8 minutes ago, we see the closest star the way it was 4.4 years ago (a distance of 4.4 ly away), we see what was going on at the other end of our Milky Way galaxy over 52,000 years ago, we are NOT seeing what is going on right now because it takes the EM waves so long to get to us.

Scientists use the Hubble/Webb telescope to find objects as far away as possible, these distant objects are not only far away in space, but we are also seeing them far back in time. Hubble/Webb can look at objects that were created just after the "Big Bang"! The Hubble telescope has seen objects that were created just 400 million years after the Big Bang or what they looked like 13 billion years ago.

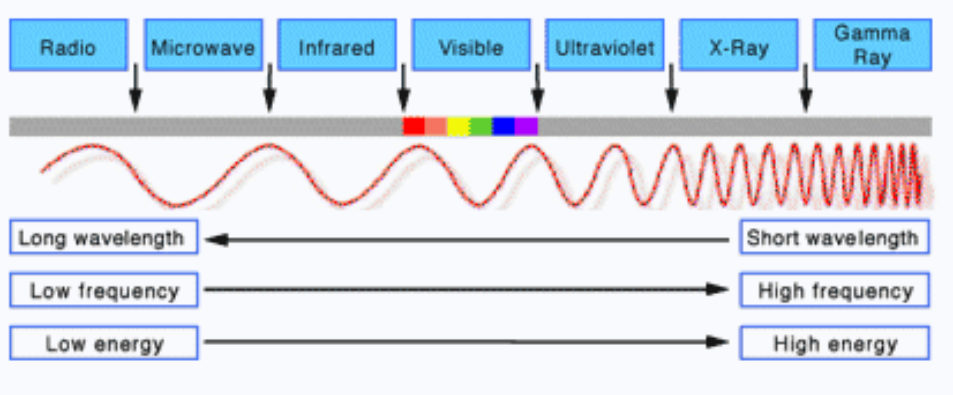


Energy in EM waves – The Energy of the electromagnetic wave is related to the frequency (f), where h is Planks Constant by

$$\textbf{\textit{Energy in EM waves}} \qquad \mathcal{E} = hf = h\frac{c}{\lambda}$$

Where h = 6.63 x 10^{-34} Js is called Planck's constant
f – frequency of wave in Hz
c – speed of light in vacuum c = 3 x 10^{8} m/s
λ – wavelength of wave in meters (m)

- The higher the frequency of the EM wave the higher its energy



Wave vs. Particle (Dual nature of light) - There was for many years a debate in the Physics community if light was a wave or a particle. Light acts like a wave when it interferes with itself and many of its properties can be explained by its wave nature, but not all. The photoelectric effect was the first experimental evidence that Light acted like a particle. Light has both properties (wave and particle) depending on how and what it interacts with and how the light is being measured.

Light can behave like both a particle and a wave!

Quantized – Restricted to a discrete quantity of energy or mass, it can only have that energy or a multiple of that energy or mass.

- A Quantized EM field is an EM field that is not allowed to have just any quantity of energy, the energy comes in fundamental bundles or quanta of energy and these quanta of energy add up to the total energy observed.
- These bundles of quantized energy are called photons.



Photon – A quantized packet of light, the photon is made up of energy and travel at the speed of light. They are not particles per say but energy packets that behave like a particle and a wave.

- A Photon has no rest mass, it is made up of pure energy.
- When an electron in an atom losses energy it must do so in energy levels from one of the allowed energies to another, the energy that is given up is the energy of the photon.
- The quantum jump is the exact energy given up or absorbed by the electron from the photon Energy = hf h – Plank's Constant = 6.626×10^{-34} Js
- Electromagnetic Radiation (Light) is made of quantized photon.

Example: Green light has a wavelength of $\lambda = 530$ nm, what is the energy in one photon of green light?

$$E = h\frac{c}{\lambda} = \frac{(6.63 \times 10^{-34}\text{Js})(3 \times 10^{8}\ \text{m/s}}{530 \times 10^{-9}\text{m}} = 3.75 \times 10^{-19}\text{J}$$

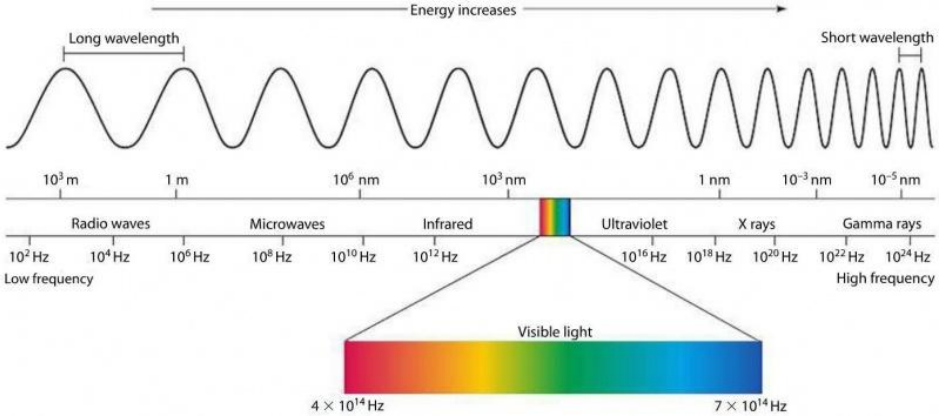
If a Laser at 530 nm produces 5 mW of green light, how many Photons are in the light each second?

$$P = \frac{\text{Enegy}}{\text{time}} = \frac{\Delta E}{\Delta t} \quad therefore\ Energy = P\Delta t = 5\ x\ 10^{-3} J\ of\ enery\ per\ second$$

$$\#\ \text{of photons} = \frac{\text{Total energy in light}}{\text{Energy per photon}} = \frac{5\ x\ 10^{-3} J}{3.75 \times 10^{-19}\ J/photon} = 1.3\ x10^{16}\ photons/s$$

Electromagnetic Spectrum – is the range of energies (or frequencies) that are possible for the electromagnetic waves. The frequency of the EM waves range from a few Hertz for radio waves to 10^{24} Hz for high energy gamma rays. The types of EM waves in the spectrum listed below are from lowest energy to highest energy.

• Radio waves • Microwaves • Infrared waves • Visible light • Ultraviolet light • X-rays • Gamma rays



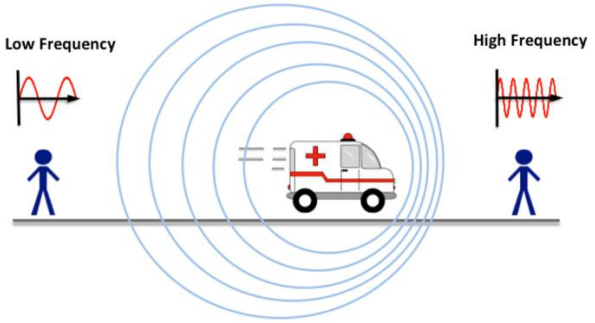
- The different parts of the spectrum of EM waves has different properties but they are part of the same spectrum.
- The different names for the different parts of the spectrum are "human-made" labels, the spectrum is continuous, it does not have breaks between the sections.
- All types of EM waves travel at the speed of light c in vacuum!
- The wavelength (λ) is another way to specify the EM wave and it is related to the Frequency in vacuum by

$$\lambda = c/f \quad \text{where} \quad c = 3 \times 10^8 \text{ m/s} \quad \text{for EM waves in vacuum}$$

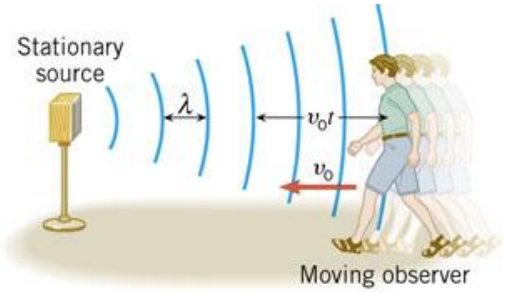
- The only part of the EM spectrum that humans can "see" is the visible section.

Doppler Effect - when a source of waves and an observer of those waves are moving relative to each other than the frequency of the waves appears to the observer to be different than that of the source if the source of the wave where stationary relative to the observer.

Moving Source – For sound waves when the source is moving the waves in front of the source bunch up because the source is moving with them and the sound waves in back of the source spread out because the source is moving away.

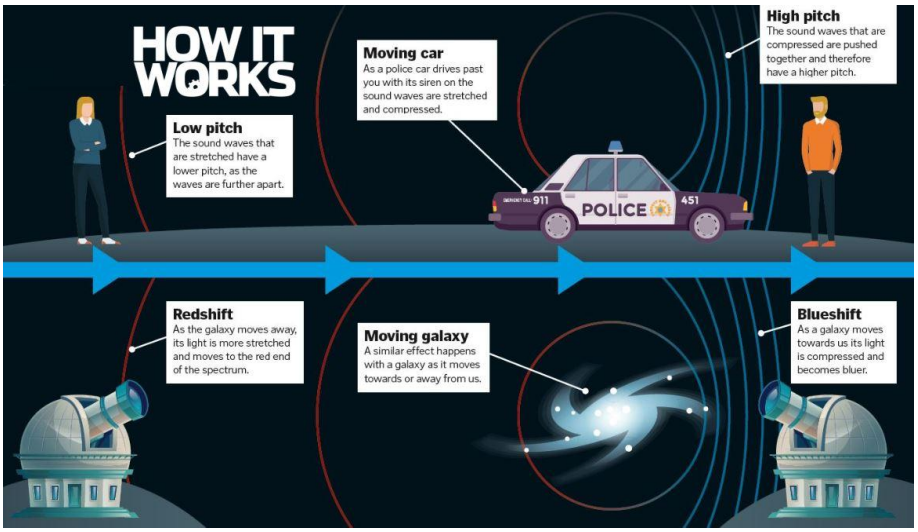


Moving Observer - Because the observer is moving the sound waves strike the observers ear at a faster rate as the observer moves towards the source and slower rate as the observer moves away from the source. The wavelength does not change in this case.



<u>Red or blue Shift</u> – refers to the shift in frequency of the EM waves from the stars and galaxy's as they move relative to the earth. For a known spectrum that is emitted by an object such as hydrogen, the spectrum is shifted due to the relative motion of the object to earth.

- Red shift - the object is moving away from earth.
- Blue shift - the object is moving towards earth.



<u>Doppler shift for electromagnetic waves</u> The Doppler shift for electromagnetic (EM) waves is the change in observed frequency (f') or wavelength due to the relative motion between a source and an observer.

$$f' = f\sqrt{\frac{1+\frac{v}{c}}{1-\frac{v}{c}}}$$

f' is the observed frequency
f is the frequency being emitted
v is the relative velocity between the source and observer
(+) Positive if moving towards and (-) negative if moving away.
c is the speed of light in vacuum.

When a source of EM waves moves relative to an observer:

- **Approaching source**: The observed frequency **increases** (blue shift).
- **Receding source**: The observed frequency **decreases** (red shift).

Applications:

- Astronomy: Used to measure the motion of stars and galaxies. A redshift indicates a galaxy is moving away, supporting the expansion of the universe.
- Radar and LIDAR: Measures speed of moving objects (e.g., vehicles).
- Medical imaging: Doppler ultrasound uses frequency shifts to measure blood flow.

Example: The red line in the hydrogen atom emits at f = 4.57 x 10^{14} Hz in a stationary reference frame. In a galaxy the same red line of hydrogen is measure to be f' = 4.34 x 10^{14} Hz, what is the relative speed of the galaxy relative to earth and in what direction?

$$f' = f\sqrt{\frac{1+\frac{v}{c}}{1-\frac{v}{c}}}$$

Solve for v

$$v = -\left[\frac{\left(\frac{f'}{f}\right)^2 - 1}{\left(\frac{f'}{f}\right)^2 + 1}\right]c = -1.54\,x\,10^7\ m/s\ \ and\ moving\ away\ (negative\ sign)$$

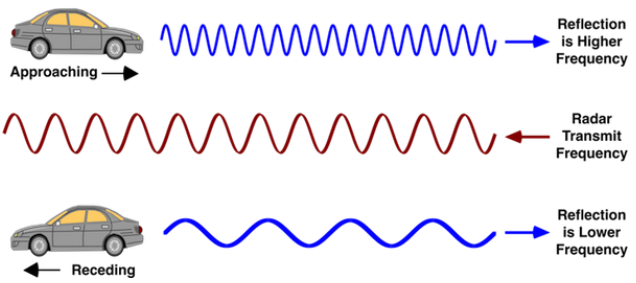
Frequency shift on Reflection from a moving Object – When a wave (sound or light) hits a moving object the reflected wave's frequency is change and this change depends on the speed of the object.

- ✓ If the object is moving towards the source of the signal, then the frequency is increased.
- ✓ If the object is moving away from the source of the signal, then the frequency is decreased.

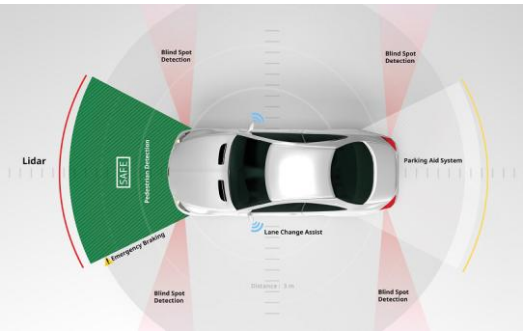
$$\Delta f = \pm 2f_s\left[\frac{v_{oject}}{v_{wave}}\right]$$

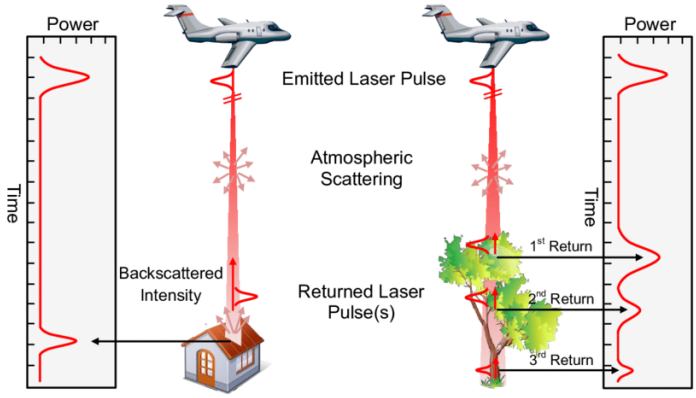
$$v_{object} = \pm\frac{\Delta f v_{wave}}{2f_s}$$

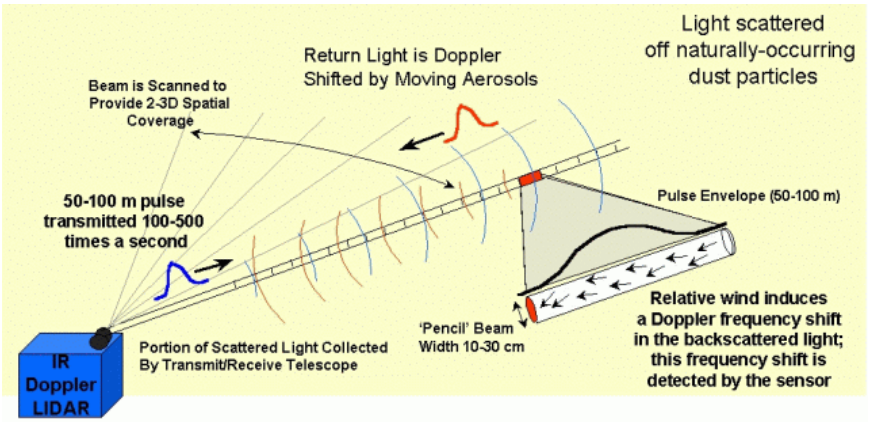
Example is the radar gun - used by police to detect the cars speed, the gun emitters a wave at a known frequency (f_s) and speed (v_{wave}), the reflected wave changes frequency relative to the gun due to the moving car (car acts as the source), the change of frequency is measured from the emitted wave and the reflected wave. This change of frequency is proportional to the speed of the car.



LIDAR – Light Detection and Ranging – Uses EM waves in the infrared and visible spectrum to detect and determine the distance to objects by sending out pulses of EM waves and analyzing the reflected waves in time and shape.



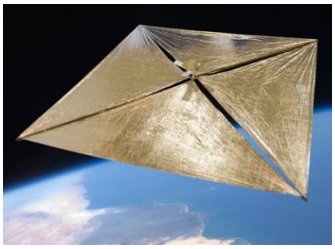




Momentum of Light (p_{light}) - Although EM waves do not have mass they do have momentum and can impart that momentum to other objects

$$p_{light} = \varepsilon_o E^2/c = B^2/\mu_o c = hf/c$$

Example: a solar sail uses light to move the sail in outer space, it can be used to transport spaceships without using rockets or fuel. When light hits the sail, it imparts its momentum to the sail much like tiny marbles hitting the sail would, but light is not made of matter but of energy so the momentum it transfers is related to the energy (hf) of the EM radiation.

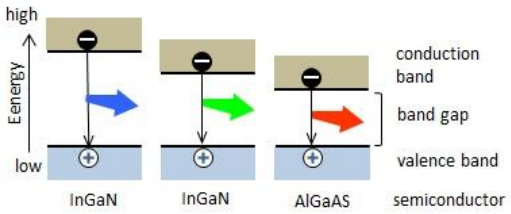


Radiation Pressure – The force per unit area of EM waves exerts on a surface due to the interaction of the surface and the EM wave (reflection or absorption).

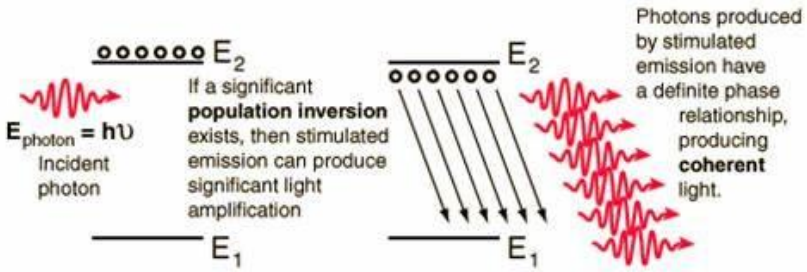
- Sunlight creates Light pressure of approximately 5×10^{-6} N/m^2

Light Emitting Diode (LED) – A semiconductor device that uses an energy gap created in two materials to produce light when a current (charge) losses energy as it goes through the gap.

- Because the energy gap creates a narrow band of colored light, the majority of the electrical energy goes into the light. Unlike incandescent bulbs where a large portion of the energy is thermal energy or heat is wasted.



LASER - **L**ight **A**mplification by **S**timulated **E**mission of **R**adiation. Lasers create light that is coherent ($\vec{E}$ fields are in phase) and monochromatic (single frequency or color). They use a material (Lasing medium) where the electrons in the atoms that have been excited into a single higher energy band. There are more simulated electrons in this excited band than the lower energy levels, this creates a population inversion of excited electrons. By producing a feedback loop (reflection) in a cavity, each photon that goes through the material will stimulate many more at the same energy and phase. As photons of light pass through these atoms the excited elections are stimulated into dropping to the lower state thereby giving up their energy in phase (coherent) with the simulating light.



This is why laser light is different than regular light sources or incoherent sources. Incoherent light the $\vec{E}$ fields are random and do not add constructively. Therefore, incoherent sources are much less intense than the laser light.

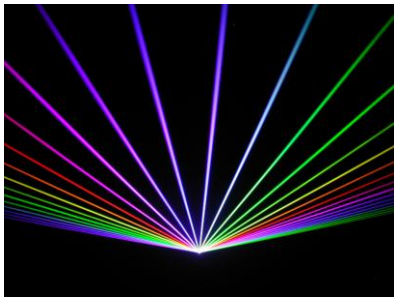
Properties of Laser Light

Monochromatic – single frequency or color.

- Incoherent sources tend to emit light over a broad frequency of color
- Can Emit specific frequencies across the entire EM spectrum.

Collimated Light - Light rays are nearly parallel and can travel long distances without significant divergence.

- Incoherent sources are difficult to collimate.



Coherent & high power – EM wave is in phase; therefore, they add together to produce very large field strengths, this is why Lasers are so "powerful."

- Incoherent sources, the EM waves are not in phase and are randomly oriented therefore not constructively.

Applications for Lasers:

Medicine

- Surgery and eye procedures: Used for precise surgeries, such as LASIK for vision correction and removing cataracts or skin lesions.
- Diagnostics: Fiber-optic endoscopies use lasers to aid in detecting issues like stomach or intestinal ulcers.
- Other treatments: Can be used to break down kidneys or gallstones and stop bleeding.

Manufacturing

- Cutting, welding, and drilling: High-powered lasers precisely cut, weld, and drill various materials like metal, plastic, and wood.
- Engraving and marking: Used to create permanent designs, logos, or text on products.
- Material processing: Includes laser cleaning, hardening, and surface modification.
- Electronics: Essential in manufacturing semiconductors for processes like photolithography and trimming integrated circuits.

Telecommunications

- Fiber optics: Lasers transmit data as pulses of light through fiber optic cables at high speeds, forming the backbone of the internet and telecommunications.
- Data storage and scanning
- Optical discs: Used to read and write data on CDs, DVDs, and Blu-ray discs.
- Barcode scanners: Lasers read barcodes on products at checkout counters.
- Laser printers: Play a key role in creating high-resolution printing plates and scanning images.
- Science and other applications

Research:

- Employed in various scientific fields, including spectroscopy, which studies the interaction of matter with light.

Measurement and guidance:

- Used in surveying equipment like laser levels and for applications like LIDAR.
- Used for target marking and in missile defense systems.